

Rapport 1 - Architecture complete du projet Metrics/GP/MCR

Date de generation: 2026-06-02 12:24:03

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2. Vue d'ensemble du systeme

L'application est une plateforme full-stack composee d'un backend FastAPI, d'un frontend React/Vite et d'une base SQL (PostgreSQL en cible, SQLite en developpement). Le coeur metier repose sur une architecture de snapshots quotidiens: chaque import MCR conserve l'etat brut des credits dans `loans_raw`, puis produit des indicateurs agreges exploites par le dashboard, les objectifs et les bonus.

3. Diagrammes

3.1 Diagramme de classes

Classes principales (ORM):

Agency (1) ---- (N) Agent

Agency (1) ---- (N) Target

Agent (1) ---- (N) Target (AGENT scope)

User (N) ---- (1) Agency [optionnel]

User (N) ---- (1) Agent [optionnel]

ImportBatch (1) ---- (N) LoanRaw

LoanRaw -> alimente calculs -> DailyMetric

BonusRule (1) ---- (N) BonusResult

User (1) ---- (N) BonusResult

3.2 Diagramme des modules

Flux modules:

- frontend/src/main.jsx
 - > frontend/src/api.js
 - > backend/app/api/v1/*
 - > backend/app/services/*
 - > backend/app/models/entities.py
 - > backend/app/db/session.py (SQL)

3.3 Schema backend/frontend/base de donnees

Flux MCR & analytics:

Fichier Excel MCR

- > api/v1/imports.py
- > services/imports.py (parse/validate/upsert + ImportBatch)
- > loans_raw (snapshot conserve)
- > services/metrics.py / api/v1/metrics.py
- > daily_metrics + dashboard KPIs/charts/tables
- > objectifs (targets.py) & bonus (bonus.py)

4. Cartographie complete des dossiers

Dossier	Fonction	Rôle dans le workflow
backend	FastAPI backend root. Contains API, services, data models and scripts.	Serves business logic, import pipeline, metrics computation, security, reporting.
backend/app	Application package root.	Wires all backend layers and exposes main app endpoint.
backend/app/api	HTTP API namespace.	Receives requests and delegates to service/model layers.
backend/app/api/v1	Versioned REST endpoints.	Entry point for auth, metrics, targets, imports, bonus, users, lookups, reports.
backend/app/assets	Supporting subdirectory for module organization.	Belongs to Project Root / Runtime Artifact and participates in that workflow.
backend/app/assets/fonts	Supporting subdirectory for module organization.	Belongs to Project Root / Runtime Artifact and participates in that workflow.
backend/app/core	Core settings/security/logging modules.	Provides JWT, configuration and logging consumed by all backend modules.
backend/app/db	Database engine/session/bootstrap utilities.	Creates SQLAlchemy engine/session and migration-like schema alignment.
backend/app/models	SQLAlchemy ORM entities and enums.	Defines persistent domain structure used by API/services.
backend/app/schemas	Pydantic request/response contracts.	Validates payloads and shapes JSON exchanged with frontend.
backend/app/services	Business services.	Implements import ETL, metrics, bonus, scope rules and objective calculations.
backend/logs	Runtime logs location.	Stores application logs for diagnostics and auditing.
backend/sample_data	Sample MCR-like files used for testing and demonstration.	Reference payloads for validating import and metrics workflows.
backend/scripts	Operational automation scripts.	Seed, backfill, sample file generation and QA/report generation.
backend/tests	Supporting subdirectory for module organization.	Belongs to Project Root / Runtime Artifact and participates in that workflow.
docs	Project documentation root.	Holds setup/API/architecture docs and generated QA reports.
docs/reports	Generated QA/test artifacts.	Contains the unique consolidated validation outputs ready for QA sharing.
docs/reports/final	Supporting subdirectory for module organization.	Belongs to Documentation/QA Artifacts and participates in that workflow.
docs/reports/final/captures	Supporting subdirectory for module organization.	Belongs to Documentation/QA Artifacts and participates in that workflow.
frontend	React frontend root.	Hosts SPA, API client and styles used by business users.
frontend/src	Frontend source code.	Implements dashboard, role-based views, filters, tables, charts and forms.
logs	Supporting subdirectory for module organization.	Belongs to Project Root / Runtime Artifact and participates in that workflow.

5. Cartographie complete des fichiers

Fichier	Fonction	Relations / role general
.env	Supporting project file.	Associated with Project Root / Runtime Artifact.
.env.example	Supporting project file.	Associated with Project Root / Runtime Artifact.
.gitignore	Supporting project file.	Associated with Project Root / Runtime Artifact.
README.md	Top-level project overview and quick-start commands.	Entry guide for developers/operators before running backend/frontend.
backend/.env	Supporting project file.	Associated with Project Root / Runtime Artifact.
backend/.env.example	Supporting project file.	Associated with Project Root / Runtime Artifact.
backend/Dockerfile	Backend container build recipe.	Packages backend app and runtime for docker-compose deployment.
backend/Dockerfile.dev	Supporting project file.	Associated with Project Root / Runtime Artifact.
backend/app/__init__.py	Supporting project file.	Associated with Project Root / Runtime Artifact.
backend/app/api/deps.py	Reusable FastAPI dependencies (auth, pagination, role checks).	Injected by endpoint modules to enforce access and query standards.
backend/app/api/v1/__init__.py	Supporting project file.	Associated with Backend API Layer.
backend/app/api/v1/auth.py	Authentication endpoints.	Validates credentials and returns JWT access tokens.
backend/app/api/v1/bonus.py	Bonus rule and execution endpoints.	Exposes formula tokens and runs monthly bonus computation.
backend/app/api/v1/complaints.py	Supporting project file.	Associated with Backend API Layer.
backend/app/api/v1/imports.py	Import endpoints for current state and historical month files.	Delegates Excel parsing/upsert/recalculation to services/imports.py.
backend/app/api/v1/lookups.py	Agency/agent lookup endpoints.	Feeds filter controls and form selectors in frontend.
backend/app/api/v1/metrics.py	Metrics, charts, snapshots and portfolio-performance endpoints.	Computes dashboard values from LoanRaw/ImportBatch with role-aware scope filters.
backend/app/api/v1/reports.py	Excel/PDF export endpoints.	Exports DailyMetric data under role/data-scope restrictions.
backend/app/api/v1/router.py	Central API router registry.	Controls which endpoint modules are exposed (e.g. complaints removed here).
backend/app/api/v1/targets.py	Objective CRUD with strict role permissions and anti-duplicate checks.	Handles agency/agent objective workflows and validates scope integrity.
backend/app/api/v1/users.py	Super Admin user management endpoints.	Creates/updates/disables users and triggers auto agent account creation.
backend/app/assets/fonts/arial.ttf	Supporting project file.	Associated with Project Root / Runtime Artifact.
backend/app/assets/fonts/arialbd.ttf	Supporting project file.	Associated with Project Root / Runtime Artifact.
backend/app/assets/fonts/tahoma.ttf	Supporting project file.	Associated with Project Root / Runtime Artifact.
backend/app/assets/fonts/tahomabd.ttf	Supporting project file.	Associated with Project Root / Runtime Artifact.
backend/app/core/config.py	Environment-backed settings model.	Supplies DB URL, CORS, JWT config to app/session layers.
backend/app/core/logging.py	Logging formatter/handler setup.	Enables file + console observability.
backend/app/core/security.py	Password hashing + JWT issuance/verification utilities.	Used by auth endpoints and dependency guards.

Fichier	Fonction	Relations / role general
backend/app/db/base.py	Supporting project file.	Associated with Database Infrastructure.
backend/app/db/migrations.py	Runtime additive schema reconciler.	Ensures missing columns/constraints are added on startup.
backend/app/db/session.py	SQLAlchemy engine/session factory.	Every endpoint/service uses this for DB access lifecycle.
backend/app/main.py	FastAPI application entryptoint.	Creates app, configures middleware/handlers, mounts API router.
backend/app/models/__init__.py	Supporting project file.	Associated with Data Model Layer.
backend/app/models/entities.py	ORM class definitions for all domain tables.	Source of truth for data relations, constraints and persisted attributes.
backend/app/models/enums.py	Enum definitions (roles, batch types and target types).	Referenced by models, schemas and permission logic.
Fichier	Fonction	Relations / role general
backend/app/schemas/auth.py	Supporting project file.	Associated with Validation/Serialization Layer.
backend/app/schemas/common.py	Supporting project file.	Associated with Validation/Serialization Layer.
backend/app/schemas/domain.py	Main API schema contracts (metrics, charts, targets, bonus payloads).	Shapes frontend payloads and computed ratio fields.
backend/app/schemas/imports.py	Import row and batch/result schemas.	Validates ETL rows and response payloads for import endpoints.
backend/app/services/agent_identity.py	Supporting project file.	Associated with Backend Service Layer.
backend/app/services/auth_security.py	Supporting project file.	Associated with Backend Service Layer.
backend/app/services/bonus.py	Formula evaluator + bonus result persistence.	Combines target values and realized metrics for payout computation.
backend/app/services/data_scope.py	Supporting project file.	Associated with Backend Service Layer.
backend/app/services/imports.py	MCR ETL core: parse/validate/upsert/import-batch logic.	Feeds LoanRaw + ImportBatch then triggers metrics recalculation and account sync.
backend/app/services/lookups.py	Supporting project file.	Associated with Backend Service Layer.
backend/app/services/metrics.py	DailyMetric recalculation service from raw loans.	Maintains aggregate table used by exports and tabular views.
backend/app/services/objective_achievement.py	Achievement-rate direction rules (higher-better vs lower-better).	Aligns PAR objective scoring with inverse achievement logic.
backend/app/services/objective_metrics.py	Objective-period realization calculations.	Used by targets follow-up and bonus calculations for month/year scopes.
backend/app/services/user_accounts.py	Supporting project file.	Associated with Backend Service Layer.
backend/dev_microcred.db	SQLite database artifact.	Local runtime storage used during development/testing scripts.
backend/logs/microcred.log	Runtime log artifact.	Captures server execution output for diagnostics.
backend/microcred.db	SQLite database artifact.	Local runtime storage used during development/testing scripts.
backend/requirements.txt	Python dependencies list.	Defines backend runtime libraries (FastAPI, SQLAlchemy, pandas, reportlab, etc.).
backend/sample_data/MCR_09052026.xlsx	Spreadsheet sample/input artifact.	Used by import workflows to populate LoanRaw snapshots.
backend/sample_data/loans_snapshot_2026-05-10.xlsx	Spreadsheet sample/input artifact.	Used by import workflows to populate LoanRaw snapshots.
backend/scripts/audit_dependencies.py	Supporting project file.	Associated with Operational Scripts.

Fichier	Fonction	Relations / role general
backend/scripts/backfill_snapshot_dates.py	Backfill utility for snapshot batch dates.	Repairs metadata consistency for existing snapshot batches.
backend/scripts/clean_renewal_pdf.py	Supporting project file.	Associated with Operational Scripts.
backend/scripts/create_sample_excel.py	Generates sample importable Excel file.	Supports demonstrations and import validation drills.
backend/scripts/generate_consolidated_validation_pdf.py	Supporting project file.	Associated with Operational Scripts.
backend/scripts/generate_env_structure_report.py	Supporting project file.	Associated with Operational Scripts.
backend/scripts/generate_final_validation_reports.py	Supporting project file.	Associated with Operational Scripts.
backend/scripts/generate_project_reports_pdf.py	Documentation report generator (architecture + user guide PDFs).	Produces governance-ready project documentation artifacts.
backend/scripts/generate_qa_report.py	Automated QA test runner + HTML/JSON report generator.	Executes API/logic checks and archives evidence per run.
backend/scripts/prepare_deployment.py	Supporting project file.	Associated with Operational Scripts.
backend/scripts/run_load_test.py	Supporting project file.	Associated with Operational Scripts.
backend/scripts/run_security_assessment.py	Supporting project file.	Associated with Operational Scripts.
backend/scripts/seed.py	Initialization script for admin + defaults + cleanup of legacy demo agencies.	Invoked by docker-compose backend startup before serving API.
backend/tests/test_security_controls.py	Supporting project file.	Associated with Project Root / Runtime Artifact.
dev_microcred.db	SQLite database artifact.	Local runtime storage used during development/testing scripts.
Fichier	Fonction	Relations / role general
docker-compose.prod.yml	Supporting project file.	Associated with Project Root / Runtime Artifact.
docker-compose.yml	Multi-service orchestration (db, backend, frontend).	Bootstraps end-to-end environment and startup commands.
docs/api.md	API-oriented documentation.	Describes endpoint contracts and usage conventions.
docs/architecture.md	Existing architecture notes.	Reference narrative that complements generated architecture PDF.
docs/reports/final/README_QA_FINAL.md	Supporting project file.	Associated with Documentation/QA Artifacts.
docs/reports/final/architecture_complete_final.pdf	Supporting project file.	Associated with Documentation/QA Artifacts.
docs/reports/final/captures/login.png	Supporting project file.	Associated with Documentation/QA Artifacts.
docs/reports/final/guide_utilisation_final.pdf	Supporting project file.	Associated with Documentation/QA Artifacts.
docs/reports/final/load_test_results.json	Supporting project file.	Associated with Documentation/QA Artifacts.
docs/reports/final/rapport_charge.md	Supporting project file.	Associated with Documentation/QA Artifacts.
docs/reports/final/rapport_securite.md	Supporting project file.	Associated with Documentation/QA Artifacts.
docs/reports/final/rapport_tests_fonctionnels.md	Supporting project file.	Associated with Documentation/QA Artifacts.
docs/reports/final/rapport_validation_final.pdf	Supporting project file.	Associated with Documentation/QA Artifacts.
docs/reports/final/structure_detaillee_projet.md	Supporting project file.	Associated with Documentation/QA Artifacts.

Fichier	Fonction	Relations / role general
docs/reports/final/validation_results.json	Supporting project file.	Associated with Documentation/QA Artifacts.
docs/setup.md	Environment setup guide.	Step-by-step local/deployment startup instructions.
frontend/Dockerfile	Frontend container build recipe.	Packages Vite app for containerized execution.
frontend/Dockerfile.dev	Supporting project file.	Associated with Project Root / Runtime Artifact.
frontend/index.html	Supporting project file.	Associated with Project Root / Runtime Artifact.
frontend/nginx.conf	Supporting project file.	Associated with Project Root / Runtime Artifact.
frontend/package-lock.json	Supporting project file.	Associated with Project Root / Runtime Artifact.
frontend/package.json	Frontend package metadata and scripts.	Defines build/dev entry commands for React/Vite app.
frontend/src/api.js	Frontend API client wrappers.	Centralizes HTTP calls to backend endpoints and auth header logic.
frontend/src/main.jsx	Single-page React application containing all feature screens.	Implements dashboard, filters, charts, import flows, bonus, targets and users.
frontend/src/styles.css	Global UI styling and responsive rules.	Controls desktop/tablet/mobile behavior, table/charts layout and touch ergonomics.
frontend/vite.config.js	Vite bundler configuration.	Build/dev server behavior for frontend assets.
logs/microcred.log	Runtime log artifact.	Captures server execution output for diagnostics.
validation_import.db	SQLite database artifact.	Local runtime storage used during development/testing scripts.

6. Fichiers critiques par workflow metier

Workflow	Fichiers responsables
Import MCR (etat actuel / mois passe / snapshots)	backend/app/api/v1/imports.py, backend/app/services/imports.py, backend/app/models/entities.py (ImportBatch, LoanRaw). Inclut la suppression securisee de sn
Calcul metriques dashboard	backend/app/api/v1/metrics.py, backend/app/services/metrics.py, backend/app/services/objective_metrics.py
Dashboard frontend	frontend/src/main.jsx, frontend/src/api.js, frontend/src/styles.css
Gestion des roles & scopes	backend/app/api/deps.py, backend/app/services/data_scope.py, backend/app/models/enums.py, frontend/src/main.jsx
Objectifs agence/agent	backend/app/api/v1/targets.py, backend/app/models/entities.py (Target), frontend/src/main.jsx
Bonus	backend/app/api/v1/bonus.py, backend/app/services/bonus.py, frontend/src/main.jsx
Exports Excel/PDF	backend/app/api/v1/reports.py, frontend/src/main.jsx (ReportButtons)
Generation rapports QA	backend/scripts/generate_qa_report.py, docs/reports/*

7. Notes sur exclusions techniques

Pour conserver un rapport lisible, l'inventaire detaille exclut les repertoires de dependances et de build (node_modules, dist, .env, __pycache__). Ces repertoires ne contiennent pas de logique metier proprietare.